SAHAYAK | SCIENCE EXHIBITION
BARAKHA DOLEY
SYSTEM DESIGN, TECHNOLOGY & WORKFLOW
ROLE-SPECIFIC SPEAKING SCRIPT
Suggested speaking time: 3-5 minutes

HANDOVER AND OVERVIEW

Thank you. I will explain how SAHAYAK works behind the screen. The core idea is to turn hand movements captured by a camera into numerical features that a trained machine-learning model can classify.

WORKFLOW

First, the browser requests camera permission. MediaPipe Hands detects hand landmarks. The application converts these points into a numerical feature vector and sends it to the Python Flask backend. The trained model predicts a label, and the website displays the result. The browser handles the camera preview and landmark overlay; the backend performs model inference.

STATIC MODEL FEATURES

Each hand has 21 landmarks, with x, y, and z coordinates for each point. That gives 21 times 3, or 63 values per hand. For two hands, the static feature vector contains 126 values: 2 x 21 x 3 = 126. The intended current labels include HELLO, YES, NO, THANK YOU, and I LOVE YOU. Recognition still depends on the model and the conditions.

SEQUENCE PATHWAY

Some signs depend on movement over time. The sequence pathway uses 132 values per frame, including landmark coordinates and wrist-position information. A sequence is resampled into 10 time bins, producing 1,320 values for the sequence model. This design represents movement over time; we should only claim a continuous sign works when it has been tested with the deployed model.

FRONTEND AND BACKEND

The frontend is hosted on Vercel and the Python Flask backend on Render. They communicate through API routes for predictions and supporting functions. Some online translation and speech services require internet access. Therefore, we must not claim that every feature works fully offline.

DATA, TRAINING AND HANDOVER

Adding a label to a list does not train a model. To add a sign properly, collect labelled examples, check data quality, train a compatible model, evaluate it on examples not used for training, and deploy the verified model. A single successful demo does not establish accuracy. My teammate will now demonstrate the prototype and explain how to test it responsibly.

JUDGE Q&A

Why use landmarks? They provide a compact numerical representation of hand shape and position.
What does 126 mean? Two hands x 21 landmarks x 3 coordinates.

Static versus sequence recognition? Static uses a frame’s features; sequence recognition uses information across time.
Is it always correct? No. Predictions can fail and must be evaluated.

REMINDERS
Do not describe the current backend as TensorFlow/CNN-based; the deployed model artifacts use scikit-learn/joblib. A feature dimension is not the number of training samples. If the demo fails, explain the workflow honestly using the architecture slide.